Lab 3: Bayesian Network for Heart Disease Diagnosis

Information Retrieval (417524B) - SPPU BE AI&DS

Write a program to construct a Bayesian Network considering medical data. Use this model to demonstrate the diagnosis of heart patients using the standard Heart Disease Data Set.

CODE

```
import warnings
warnings.filterwarnings("ignore")
import pandas as pd
import networkx as nx
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from pgmpy.models import DiscreteBayesianNetwork
from pgmpy.inference import VariableElimination

cols = ["age", "sex", "cp", "trestbps", "chol", "fbs", "restecg",
        "thalach", "exang", "oldpeak", "slope", "ca", "thal", "num"]
df = pd.read_csv(
    r"C:\Users\praka\Desktop\Academics & Coursework\be\_data\processed.cleveland.data",
    names=cols)

data = pd.DataFrame()
data["Age"] = (df["age"] > 55).map({True: "old", False: "young"})
data["Chol"] = (df["chol"] > 240).map({True: "high", False: "normal"})
data["MaxHR"] = (df["thalach"] > 150).map({True: "high", False: "low"})
data["HeartDisease"] = (df["num"] > 0).map({True: "yes", False: "no"})
# data["Chol"] = (df["chol"] > 200).map({True: "high", False: "normal"})
# data["Age"] = (df["age"] > 50).map({True: "old", False: "young"})
print("Discretised data (first 5 rows):")
print(data.head())

model = DiscreteBayesianNetwork([
    ("Age", "HeartDisease"),
    ("Chol", "HeartDisease"),
    ("MaxHR", "HeartDisease"),
])
model.fit(data)

infer = VariableElimination(model)
q1 = infer.query(["HeartDisease"], evidence={"Age": "old", "Chol": "high"})
q2 = infer.query(["HeartDisease"], evidence={"MaxHR": "low"})
# q3 = infer.query(["HeartDisease"], evidence={"Age": "young"})
print("\nP(HeartDisease | Age=old, Chol=high):")
print(q1)
print("\nP(HeartDisease | MaxHR=low):")
print(q2)

G = nx.DiGraph(model.edges())
nx.draw(G, nx.spring_layout(G, seed=1), with_labels=True,
        node_color="lightblue", node_size=2500, font_size=9)
plt.title("Bayesian Network for Heart Disease")
plt.savefig(
    r"C:\Users\praka\Desktop\Academics &
Coursework\be\Information_Retrieval_417524B\output\figures\lab3_bayesian_network.png",
    dpi=100)
```

OUTPUT

```
Discretised data (first 5 rows):
     Age    Chol MaxHR HeartDisease
0    old  normal   low           no
1    old    high   low          yes
2    old  normal   low          yes
3  young    high  high           no
4  young  normal  high           no

P(HeartDisease | Age=old, Chol=high):
+------------------+--------------------+
| HeartDisease     |   phi(HeartDisease) |
+==================+====================+
```

HeartDisease(no)	0.3739
HeartDisease(yes)	0.6261

P(HeartDisease | MaxHR=low):

HeartDisease	phi(HeartDisease)
HeartDisease(no)	0.3354
HeartDisease(yes)	0.6646
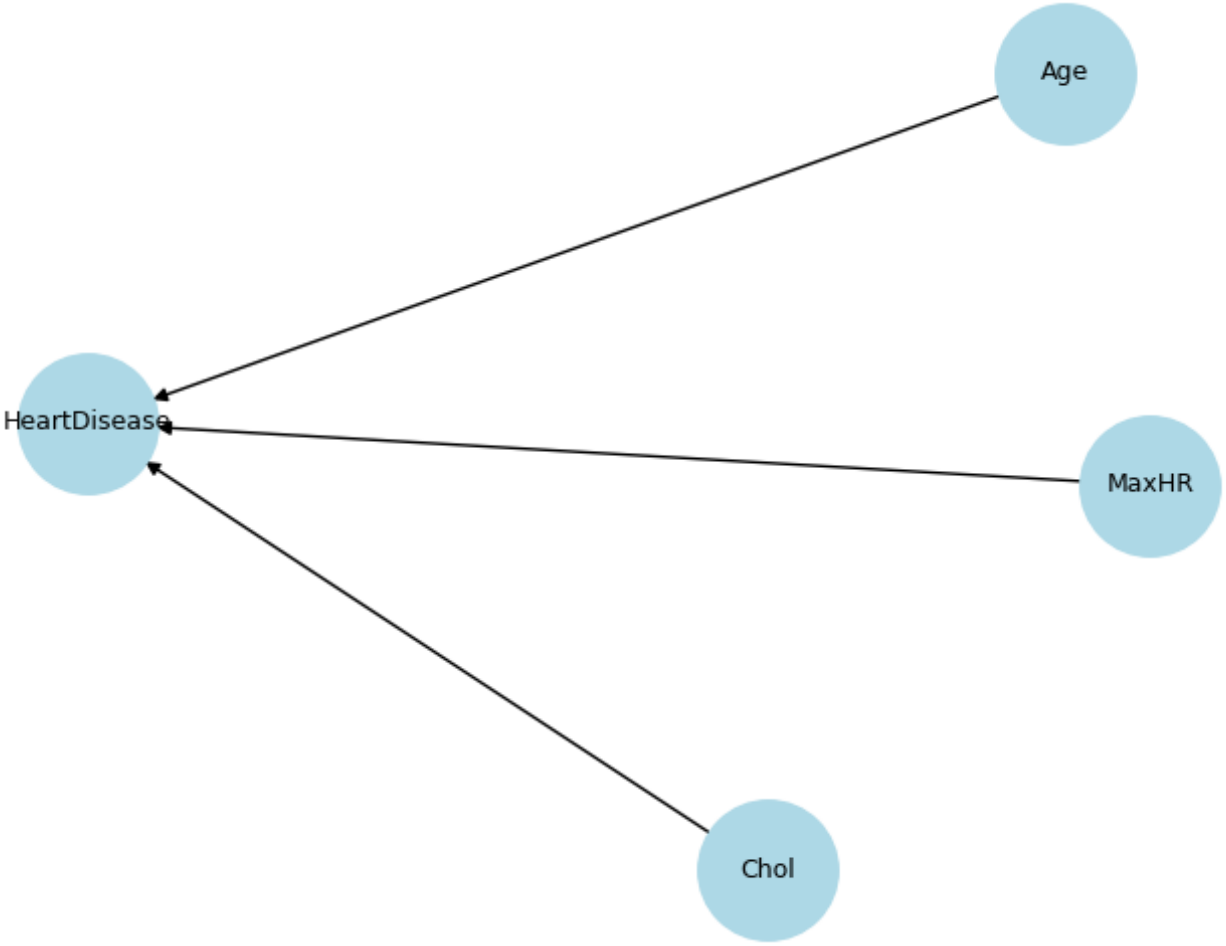
FIGURE: lab3_bayesian_network